

Technical runbook

Synthetic local demonstration. Windows operations, health, and release checks.

Getting started on Windows

Prerequisites

- Windows 10 or 11 with PowerShell
- Docker Desktop, using the WSL2 backend
- Git for Windows (`C:\Program Files\Git\cmd\git.exe` first on `PATH`)`
- `[uv](https://docs.astral.sh/uv/)` for the current user, typically `%USERPROFILE%\local\bin\uv.exe``

Do not use ``py -3.11`` for this repo. On this machine that launcher can report that the file is missing. Setup creates ``.venv`` with Python 3.11 through ``uv``.

First-time setup

From the repository root:

```
```powershell
```

```
Set-ExecutionPolicy -Scope Process Bypass
```

```
.\scripts\windows\Setup-ImpactLab.ps1
```

```
.\scripts\windows\Start-ImpactLab.ps1
```

```
```
```

``Setup-ImpactLab.ps1`` checks Docker, creates ``.venv``, installs ``requirements.txt``, creates ``.runtime\logs``, and validates ``docker compose config``. It does not change global Python settings.

Expected URLs

| Surface | URL |

|---|---|

| Dashboard | `http://127.0.0.1:8501` |

| API health | `http://127.0.0.1:8000/health` |

| API docs | `http://127.0.0.1:8000/docs` |

| PostgreSQL | ``127.0.0.1:5432``, database ``.impact_lab``, user ``.impact`` |

Verify

```
```powershell
```

```
.\scripts\windows\Status-ImpactLab.ps1
```

```
.\scripts\windows\Doctor-ImpactLab.ps1
```

```
```
```

The dashboard badge should read `SYNTHETIC DEMO` and `PostgreSQL`. Enrollment facts should be 5,760 and the seven-fact total should be 23,745 for seed `20260923`.

Daily runbook

```powershell

.\scripts\windows\Start-ImpactLab.ps1

.\scripts\windows\Status-ImpactLab.ps1

.\scripts\windows\Stop-ImpactLab.ps1

```

| Need | Command |

|---|---|

| Start without opening a browser | `.\scripts\windows\Start-ImpactLab.ps1 -NoBrowser` |

| Start and reseed | `.\scripts\windows\Start-ImpactLab.ps1 -Reseed` |

| Logs | `.\runtime\logs\api.log`, `.\runtime\logs\api.err.log`, `.\runtime\logs\dashboard.log`,
`.\runtime\logs\dashboard.err.log` |

| Stop API and dashboard only | `.\scripts\windows\Stop-ImpactLab.ps1` |

| Also stop PostgreSQL | `.\scripts\windows\Stop-ImpactLab.ps1 -StopDatabase` |

| Reseed after typing `RESEED` | `.\scripts\windows\Reset-ImpactLabData.ps1` |

| Read-only diagnosis | `.\scripts\windows\Doctor-ImpactLab.ps1` |

URLs: <http://127.0.0.1:8501> and <http://127.0.0.1:8000/docs>.

A second start while both processes are alive prints the URLs and exits. It does not launch another pair.

Operations runbook

Lifecycle

``Setup-ImpactLab.ps1`` prepares the machine-local environment. ``Start-ImpactLab.ps1`` starts PostgreSQL, seeds when enrollment is below 5,000, and starts the API and dashboard. ``Stop-ImpactLab.ps1`` stops only those recorded processes. ``-StopDatabase`` stops the Compose service and keeps the volume.

Health

Check	Command or URL
Engine	``docker version``
PostgreSQL	``pg_isready`` inside the Compose container
API	`http://127.0.0.1:8000/health`
Dashboard	`http://127.0.0.1:8501`
Bundle	``.scripts\windows\Doctor-ImpactLab.ps1``

Logs

``.runtime\logs\api.log``, ``.runtime\logs\api.err.log``, ``.runtime\logs\dashboard.log``, and ``.runtime\logs\dashboard.err.log``. ``.runtime/`` is gitignored.

Deterministic reset

``Reset-ImpactLabData.ps1`` asks you to type ``RESEED`` unless ``-Force`` is passed. It refuses to run unless ``.DATABASE_URL`` is the PostgreSQL ``.impact_lab`` database, then reruns the generator and checks that fact volume stays at least 20,000. Every row is synthetic.

Release verification

Follow [Release Checklist](RELEASE_CHECKLIST.md). CI on ``.main`` must pass the SQLite job and the PostgreSQL job.

Constraints

No production credentials, no real participant data, and no outbound chat integrations. Trust authentication on PostgreSQL is only for this localhost demo and the ephemeral CI service.

Windows troubleshooting

Format: **Symptom** → **Likely cause** → **Diagnostic** → **Safe repair** → **Validation**

Broken Python launcher

Symptom. `py -3.11`` reports that the system cannot find the file specified.

Likely cause. The `py`` launcher is stale or points at a removed install.

Diagnostic. ``Get-Command py`` and ``uv python list``.

Safe repair. Use ``uv`` to create ``.venv`` with Python 3.11. ``Setup-ImpactLab.ps1`` does this. Do not repair unrelated global Python installs.

Validation. ``.venv\Scripts\python.exe --version`` prints 3.11.

PowerShell activation blocked

Symptom. Running a script returns an execution-policy error.

Likely cause. The current user policy is Restricted.

Diagnostic. ``Get-ExecutionPolicy -List``.

Safe repair. ``Set-ExecutionPolicy -Scope Process Bypass`` in that terminal only.

Validation. ``.scripts\windows\Doctor-ImpactLab.ps1`` runs.

Hermes Git shim

Symptom. ``BUG (fork bomb)`` from a path ending in ``hermes\git\bin\git.exe``.

Likely cause. A Hermes Git shim is ahead of Git for Windows on ``PATH``.

Diagnostic. ``where.exe git``.

Safe repair. Put ``C:\Program Files\Git\cmd\git.exe`` first. Do not delete Hermes as the first step.

Validation. ``where.exe git`` lists Git for Windows first, and ``git --version`` returns.

GitHub authentication

****Symptom.**** `git push`` asks for credentials or fails.

****Likely cause.**** Git Credential Manager has no stored GitHub login.

****Diagnostic.**** `git credential-manager --version`` or `where.exe git-credential-manager``.

****Safe repair.**** Sign in through Git Credential Manager. Do not paste tokens into files or docs.

****Validation.**** `git ls-remote origin`` succeeds.

Docker and WSL

****Symptom.**** `docker version`` has a client and no server, or Compose cannot start.

****Likely cause.**** Docker Desktop is stopped, or WSL2 is not available.

****Diagnostic.**** `wsl --version``, `wsl --status``, `docker version``, `docker info``, `docker compose version``, `docker run --rm hello-world``.

****Safe repair.**** Start Docker Desktop and wait until the engine is running. Confirm the Virtual Machine Platform and WSL features only if those commands show they are missing. Do not unregister WSL or factory-reset Docker as the first fix.

****Validation.**** `docker compose config`` from the repo succeeds and `docker compose up -d`` starts PostgreSQL.

Port already in use

****Symptom.**** API, Streamlit, or PostgreSQL exits because 8000, 8501, or 5432 is taken.

****Likely cause.**** A previous lab process, or another local service.

****Diagnostic.**** `Get-NetTCPConnection -LocalPort 5432,8000,8501 -State Listen`` and `.\scripts\windows\Status-ImpactLab.ps1``.

****Safe repair.**** `.\scripts\windows\Stop-ImpactLab.ps1``. If the listener is not this project, stop that specific process. Do not stop every Python process.

****Validation.**** Status shows the expected listeners only after start.

PostgreSQL unhealthy

****Symptom.**** Start fails while waiting for PostgreSQL, or `pg\_isready` fails.

****Likely cause.**** The container is new, crashed, or Compose is not the owner of an older container.

****Diagnostic.**** `docker compose ps` and `docker compose logs postgres`.

****Safe repair.**** `docker compose up -d` from the repo. If a nameless leftover container blocks the name, remove that container only after confirming it uses the `nonprofit-impact-intelligence-lab\_pgdata` volume, then compose up again. Do not delete the volume.

****Validation.**** `docker exec nonprofit-impact-intelligence-lab-postgres-1 pg\_isready -U impact -d impact\_lab`.

Dashboard has no rows

****Symptom.**** Pages load and metrics are zero or queries fail.

****Likely cause.**** `DATABASE\_URL` points at an empty database, or seed did not run.

****Diagnostic.**** Status output and `.runtime\logs\dashboard.err.log`.

****Safe repair.**** `.\scripts\windows\Start-ImpactLab.ps1 -Reseed`, or `Reset-ImpactLabData.ps1` after typing `RESEED`.

****Validation.**** Enrollment count is at least 5,000.

API unavailable

****Symptom.**** `http://127.0.0.1:8000/health` does not answer.

****Likely cause.**** Unicorn exited, or port 8000 is busy.

****Diagnostic.**** `.\scripts\windows\Status-ImpactLab.ps1` and the tail of `.runtime\logs\api.err.log`.

****Safe repair.**** Stop the lab, free port 8000 if a foreign process owns it, then start again.

****Validation.**** `/health` returns `{"status":"ok","synthetic":true}`.

Architecture

Streamlit and FastAPI read one relational database. The generator is the only writer in this demo.

```
```mermaid
flowchart LR
 gen[Deterministic generator] --> db[(SQLite or PostgreSQL)]
 db --> api[FastAPI]
 db --> dash[Streamlit dashboard]
 api --> exports[CSV exports]
 dash --> files[CSV XLSX PDF presets]
```
```

```
```mermaid
flowchart TB
 subgraph windows [Windows scripts]
 start[Start-ImpactLab.ps1]
 end
 start --> compose[Docker Compose PostgreSQL 16]
 start --> api[Uvicorn 127.0.0.1:8000]
 start --> dash[Streamlit 127.0.0.1:8501]
 compose --> db[(impact_lab)]
 api --> db
 dash --> db
```
```

| Component | Port | Notes |
|-----------|------|-------|
|-----------|------|-------|

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|-------------|
| --- --- --- |
|-------------|

| | | |
|------------|------|--|
| PostgreSQL | 5432 | localhost only, trust auth for this demo |
|------------|------|--|

| | | |
|---------|------|-----------------|
| FastAPI | 8000 | stateless reads |
|---------|------|-----------------|

| Streamlit | 8501 | seven pages, preset exports |

SQLite remains the zero-config path when `DATABASE\_URL` is unset. There is no arbitrary SQL endpoint.